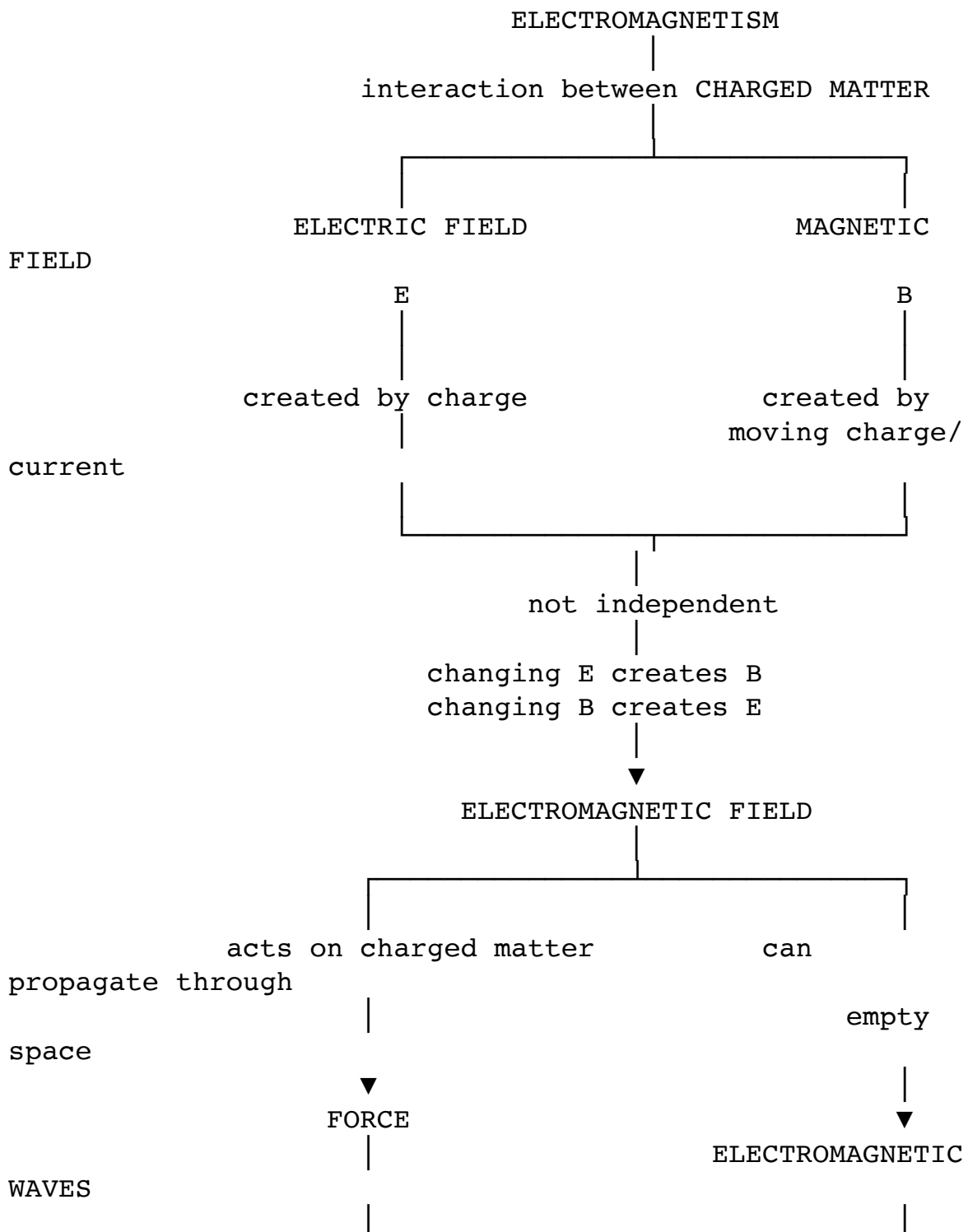
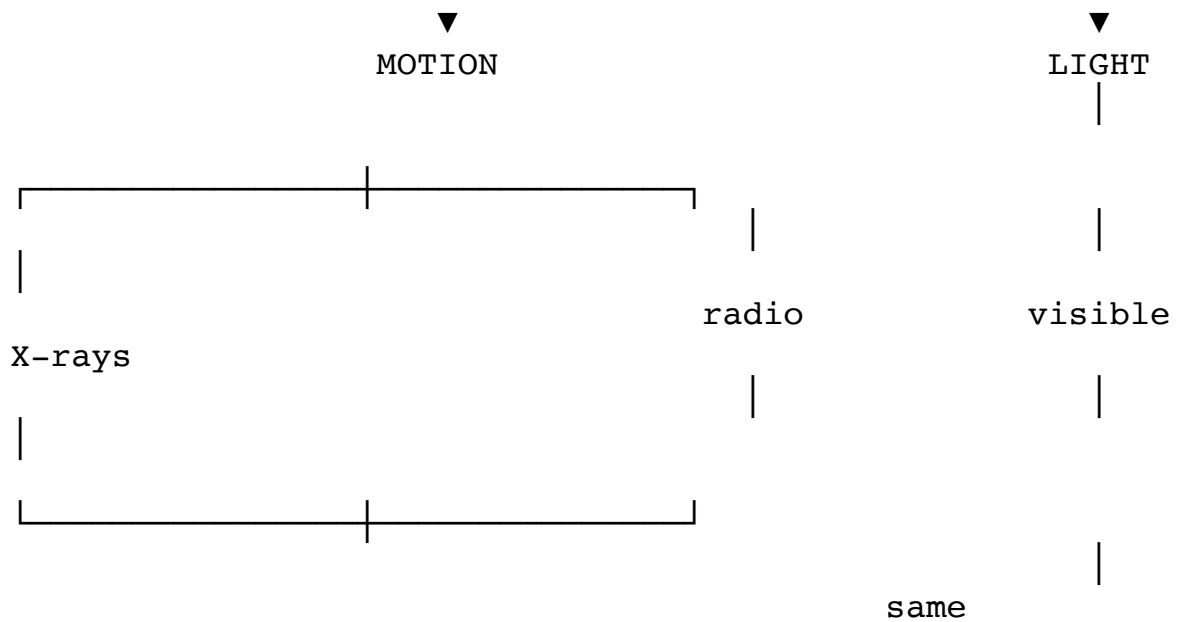


Yes. Electromagnetism is actually an excellent test case because a conventional explanation quickly becomes a linear parade of charge $\rightarrow$ fields $\rightarrow$ current $\rightarrow$ magnetism $\rightarrow$ Maxwell $\rightarrow$ photons, while the concept itself is deeply relational.

Instead of teaching it immediately, I'd first build the **map**, then use that map to decide where we need explanatory depth.

1. The Map — electromagnetism as a system





phenomenon

different

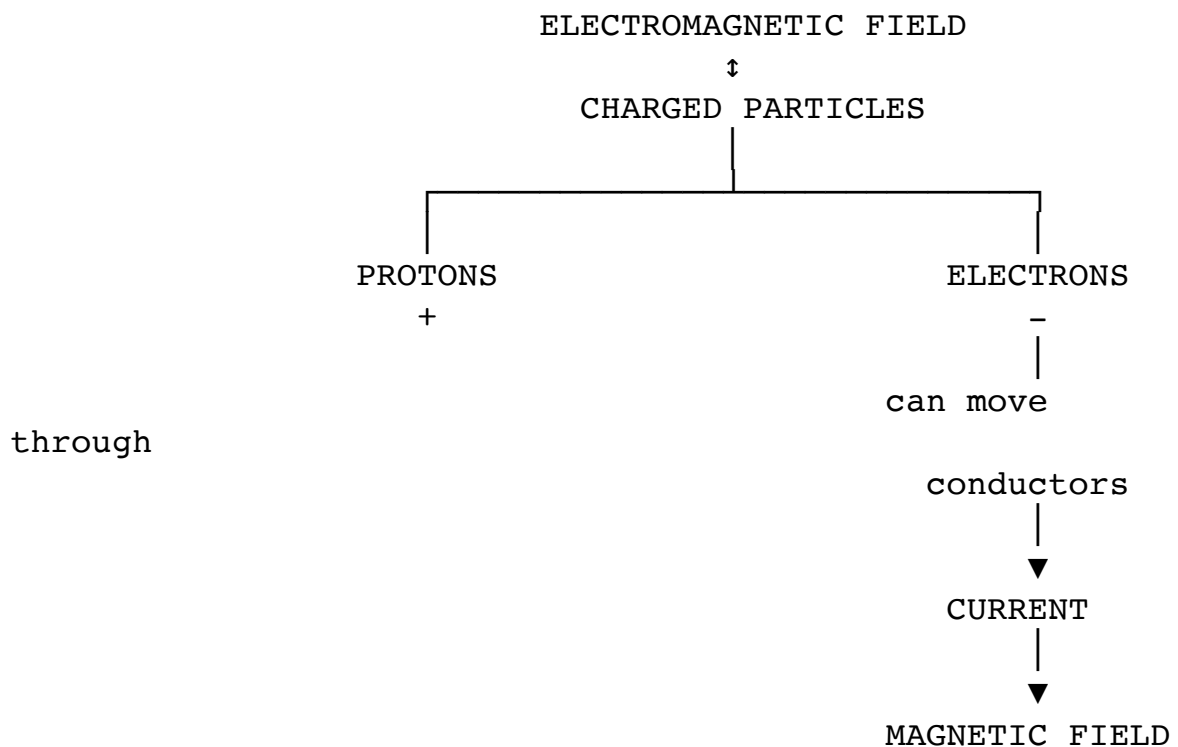
frequencies

Already, I think this communicates something that paragraphs often obscure:

Electricity, magnetism and light aren't three separate phenomena.

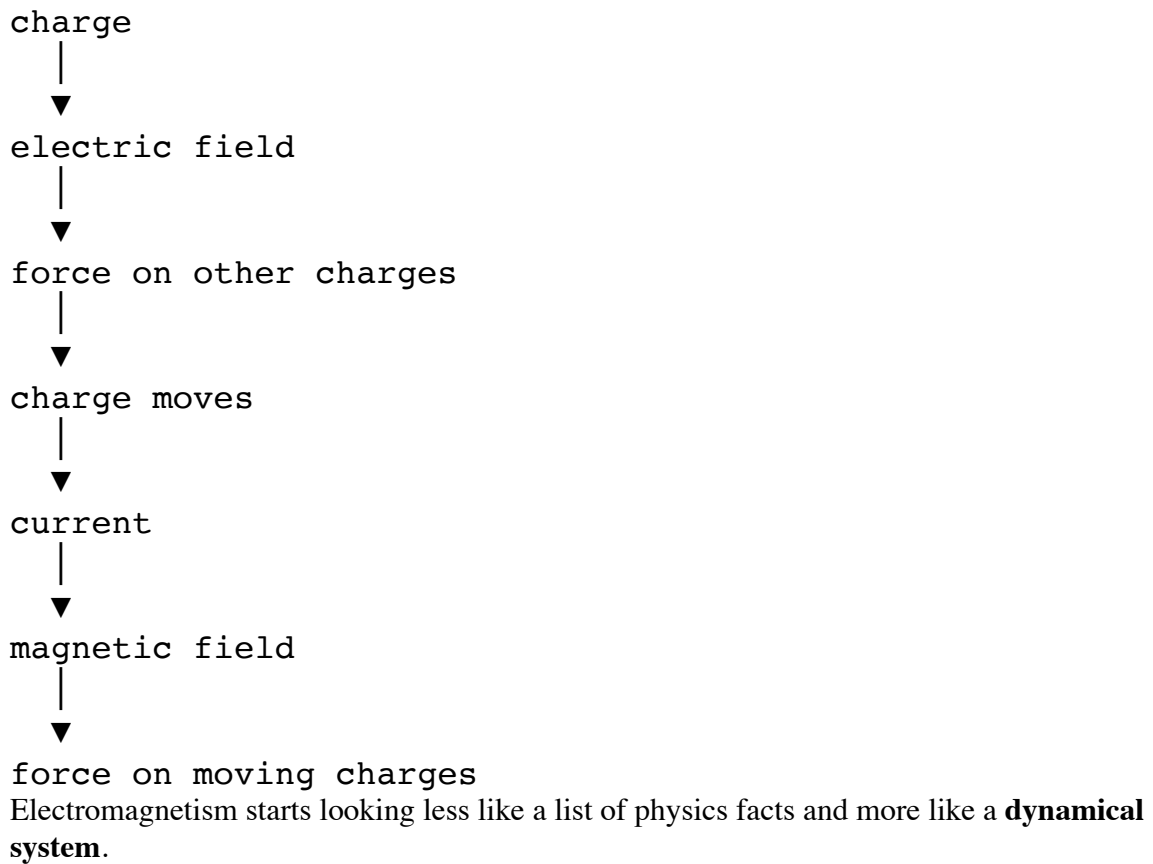
They're manifestations of **one electromagnetic system**.

Now let's add matter.



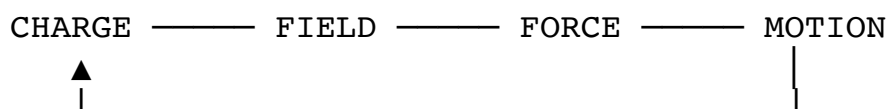
through

And now an important feedback structure emerges:



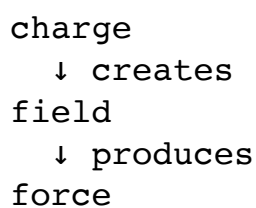
2. The Model — what actually interacts with what?

We can compress most introductory electromagnetism into four entities:



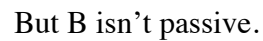
Why the loop?

Because:

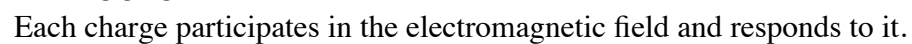


Imagine two electrons.

electron B (-)



electron B (-)

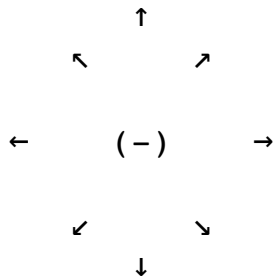


Matter changes fields; fields change matter.

Start with stationary charge.

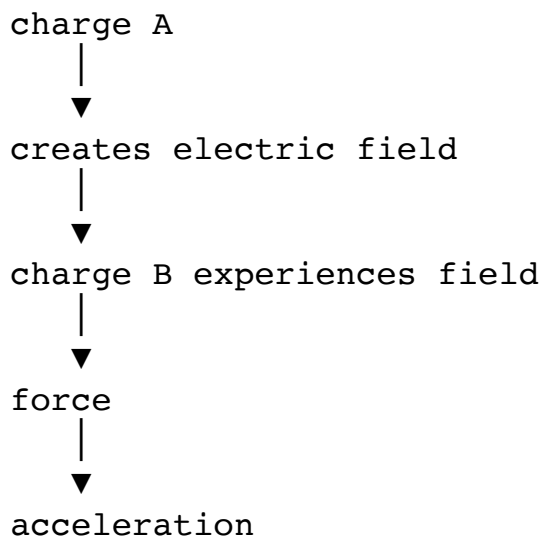
An electron produces an electric field around itself.

Conceptually:



For a negative charge, the field direction points inward; for a positive charge, outward.

Put another charge in that field:



The familiar statement:

Opposite charges attract; like charges repel.

is therefore the **observable behavior** of the deeper model.

4. Now move the charges

This is where electricity starts turning into magnetism.

Suppose electrons move through a wire:

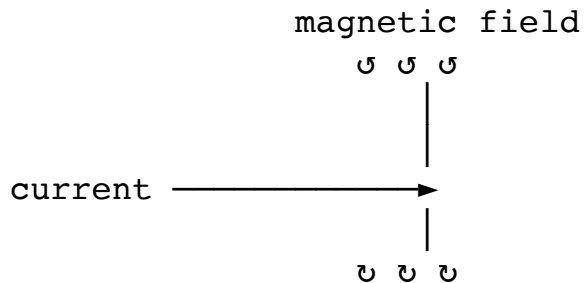
WIRE

$e^- \rightarrow e^- \rightarrow e^- \rightarrow e^- \rightarrow e^-$

electric current

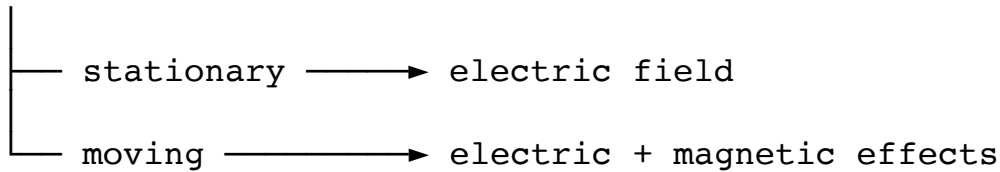
Moving charge = **electric current**.

But moving charges produce something else:



So:

CHARGE



This is our first major conceptual bridge.

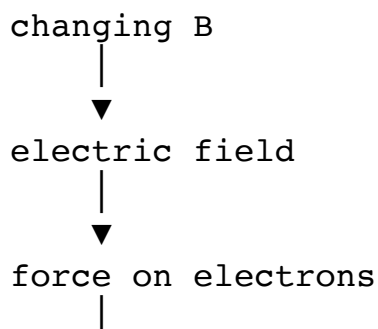
Electricity and magnetism aren't arbitrary phenomena someone decided to put in the same chapter.

They're structurally connected.

5. Now introduce change

Things get much more interesting when fields **change over time**.

A changing magnetic field generates an electric field:





current

This is electromagnetic induction.

And this seemingly abstract relationship gives us:

mechanical motion



changing magnetic field



electric field



electron movement



electric current

Which means:

generator

ROTATION → MAGNETIC CHANGE → ELECTRICITY

Reverse the relationship:

ELECTRICITY → MAGNETIC FIELD → FORCE → ROTATION

motor

Same system.

Different direction of energy conversion.

That's an example of why understanding relationships beats memorizing definitions.

6. Then comes the beautiful part

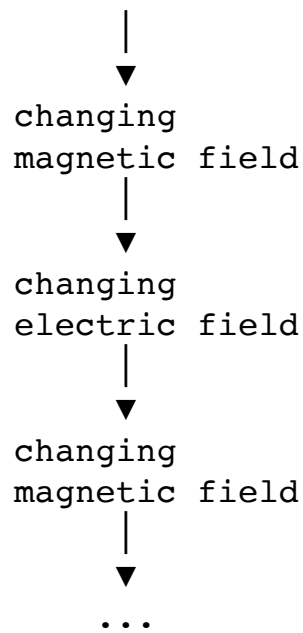
A changing magnetic field creates an electric field.

But a changing electric field also creates a magnetic field.

So imagine creating a disturbance:

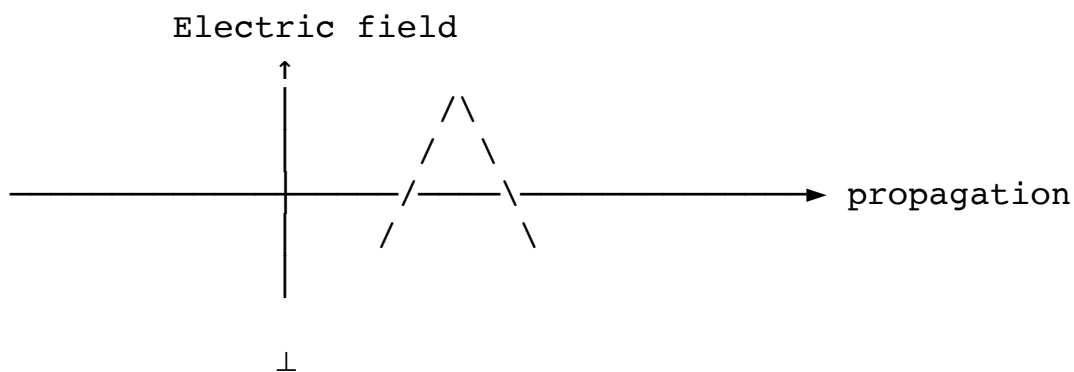
changing

electric field



The disturbance can sustain itself and propagate.

Schematically:



Magnetic field

That's an **electromagnetic wave**.

And its propagation speed in vacuum is:

$c \approx 299,792,458 \text{ m/s}$.

Which happens to be...

the speed of light.

This wasn't merely a coincidence discovered afterward. Maxwell's electromagnetic theory implied waves propagating at a speed corresponding to the known speed of light.

And suddenly:

ELECTRICITY



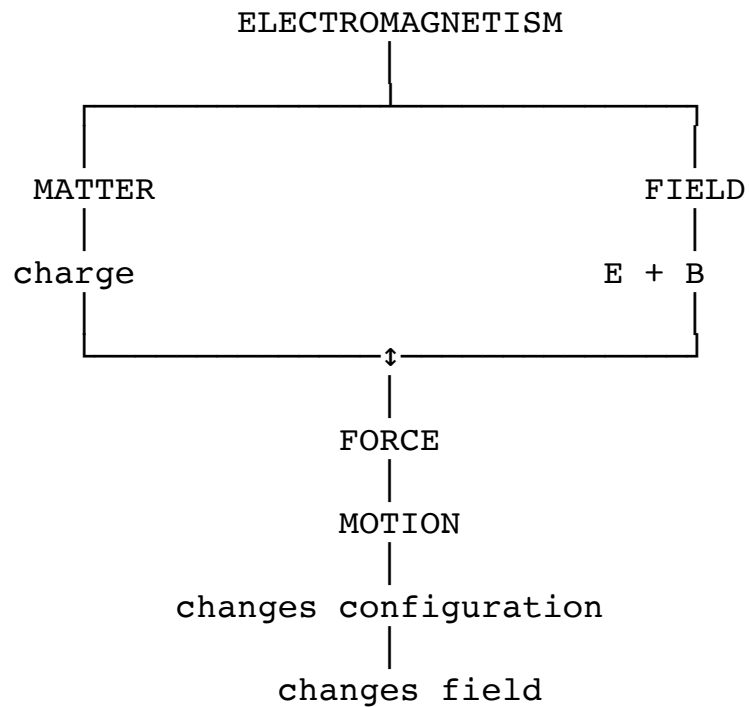
MAGNETISM

Once you've understood electromagnetic waves, the electromagnetic spectrum becomes almost trivial conceptually:

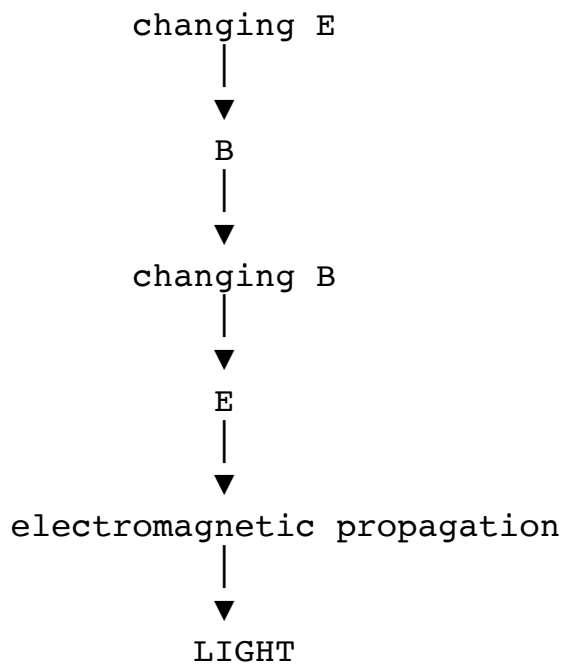
The diagram illustrates the electromagnetic spectrum as a horizontal line with tick marks. Above the line, the regions are labeled from left to right: Gamma, Radio, Microwave, Infrared, Visible, UV, and X-ray. A vertical arrow points downwards from the boundary between the Infrared and Visible regions.

8. Now zoom out one level

Our knowledge map has undergone a fairly dramatic compression:

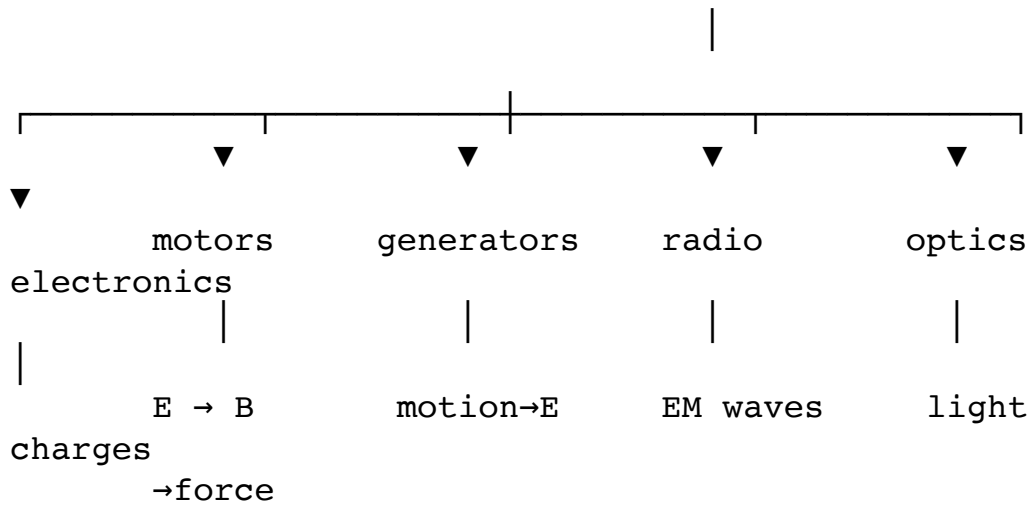


And fields themselves have dynamics:



Then practical technology attaches to particular relationships:

ELECTROMAGNETISM



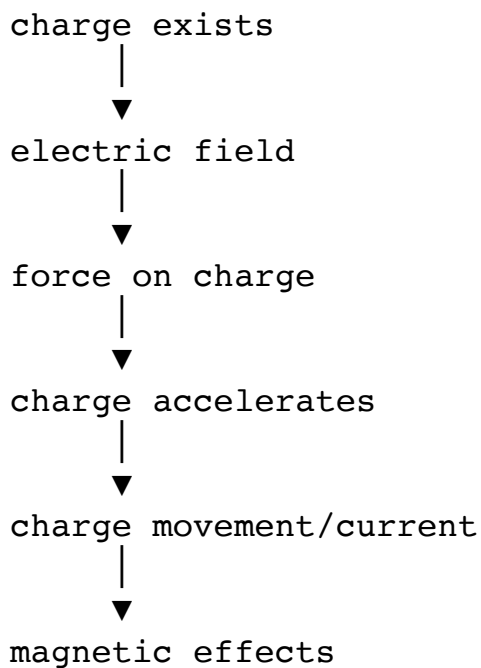
Instead of remembering five unrelated technologies, you can locate them on the underlying model.

9. But now I'd switch representation again

A map is becoming less useful because we're asking a different question:

What causes what?

So I'd generate a causal model:



And:

changing magnetic field

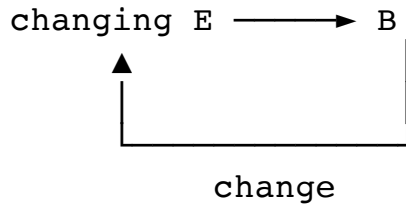


electric field



current

And:



propagating wave



light

These three small models probably teach you more than a giant mind map would.

10. Now the Simulator

This is where I'd test whether you've built an actual mental model rather than memorized my diagrams.

Don't look for memorized definitions. **Run the system forward in your head.**

Experiment A

I place a stationary electric charge in empty space.

What exists around it?

→ Electric field.

Now I introduce another charge.

→ It experiences force.

Let it go.

→ It accelerates.

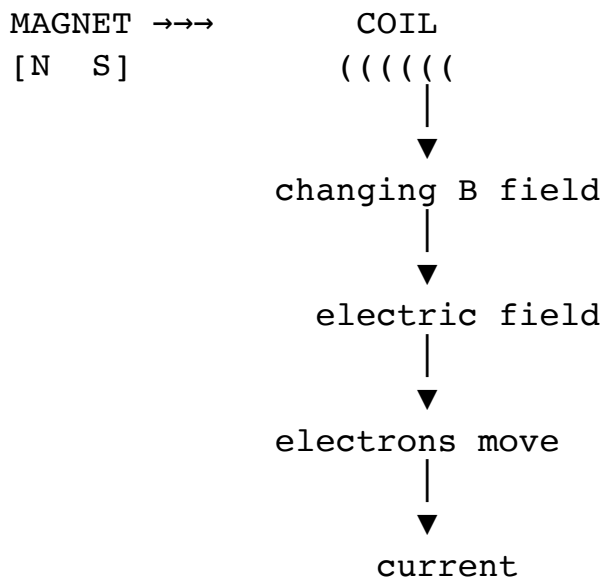
Now charges are moving.

→ Magnetic effects appear.

You've simulated the system.

Experiment B

Move a magnet through a coil of wire.



You should now be able to predict that current appears **without memorizing “Faraday’s law causes induction.”**

Experiment C

Stop moving the magnet.

magnet stationary
↓
magnetic field no longer changing
↓
induced electric field disappears
↓
induced current disappears
That’s understanding.

11. Only

now

would I introduce the mathematics

And this reveals another potential improvement to how you learn.

Instead of:

equation → explanation

I'd generally use:

SYSTEM

↓

RELATIONSHIP

↓

INTUITION

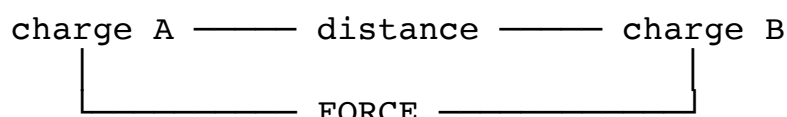
↓

MATHEMATICS

For example, once you understand that charges generate electric fields and other charges respond to them, Coulomb's law becomes a **quantification of an existing relationship**, rather than mysterious notation:

$$F = k \frac{|q_1 q_2|}{r^2}$$

The model already told us:



The equation now tells us **how much**:

more charge

↓

more force

greater distance

↓

much less force

↓

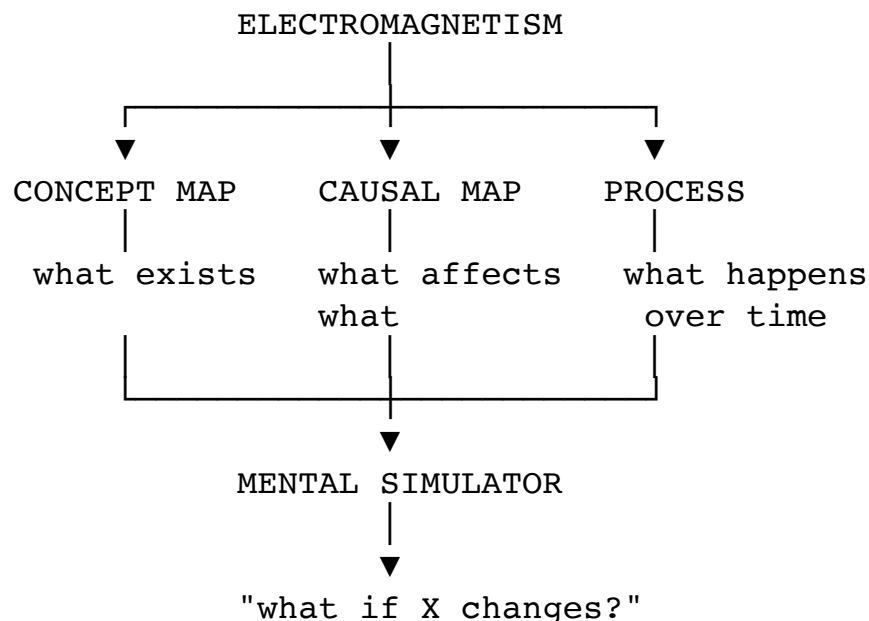
$1/r^2$

Likewise, Maxwell's equations should eventually arrive not as four equations to memorize, but as the mathematical compression of the system we've constructed.

And this experiment tells me something useful about your original question

I don't think I'd recommend **text** → **mind map** for you.

Even in this relatively simple example, I naturally wanted at least four representations:



That's much closer to what I think you're looking for.

The source can remain a textbook, paper, article or documentation. We don't need to replace text as an information-delivery medium. Instead, we insert a transformation layer between **source material** and **your mental model**.

And I think there's an especially promising rule here:

Don't visualize sentences. Extract the system described by the sentences.

That's a considerably more powerful idea than automatic mind mapping.